

Cluster Training Performance Overview — v2

Version: 2 | **Date:** 2026-04-09 **Supersedes:** cluster-training-performance-overview.md (v1, Rack 04 + Rack 11 only)

What's new in v2

- GPT-OSS 120B and Qwen 72B comparisons now cover **all four production racks** (01, 04, 11, 16).
- New Nemotron 340B TP8 run using **Rack 01 + Rack 16** (nemo340b-tp8-rack1-rack16.log) added alongside the existing Rack 04 + Rack 11 long-job reference.
- Statistics now report **median** as the primary metric (robust to isolated spikes) alongside mean for continuity with v1.
- NCCL all_reduce bandwidth sweep added (see new section below).

Executive Summary

Workload	Scope	Median step time (s)	Median TFLOPS/ GPU	Key observation
GPT-OSS 120B BF16	4 racks: 01, 04, 11, 16	5.47-5.62	395.9-407.2	Racks 04/11/16 within 0.4% of each other; Rack 01 ~2.5% lower TFLOPS
Qwen 72B BF16	4 racks: 01, 04, 11, 16	19.84-20.57	708.7-735.0	Rack 16 fastest (~3.7% above Rack 04/11); Rack 01 mid-range
Nemotron 340B TP8 — Rack 04+11	128 GPU, KAI JobSet	5.723	1475.3	545 post-warmup steps
Nemotron 340B TP8 — Rack 01+16	128 GPU, KAI JobSet	5.697	1482.0	594 post-warmup steps; +0.5% TFLOPS vs Rack 04+11

Rack-to-rack variation remains operationally small across all four racks for all tested workloads. Rack 01 shows a consistent ~2-3% throughput reduction on GPT-OSS; its Qwen 72B and Nemotron 340B results are normal. No rack shows a pattern suggesting hardware fault.

Common Method

- Extract DLL / train_step_timing / TFLOPS_per_GPU lines from each log.

- Exclude the first step (large initialization/warmup outlier present in every run) and the first 10 extracted steps as warmup normalization.
- **Primary metric: median** (robust to isolated outliers). Mean provided for continuity with v1.
- Dashed reference lines on charts: per-rack post-warmup median.
- NeMo DLL logs emit two entries per step (one with training TFLOPS, one with a low auxiliary TFLOPS value ~1.4); filter to entries with TFLOPS > 100 before computing Nemotron statistics.

GPT-OSS 120B BF16 — 4-Rack Comparison

Inputs

Rack	Log file
01	logs/prod-cluster/gpt-oss/gpt-oss-rack01-logs.txt
04	logs/prod-cluster/gpt-oss/gpt-oss-rack4-logs.txt
11	logs/prod-cluster/gpt-oss/gpt-oss-rack11-logs.txt
16	logs/prod-cluster/gpt-oss/gpt-oss-rack16-logs.txt

Results

Metric	Rack 01	Rack 04	Rack 11	Rack 16
Post-warmup steps	185	120	109	211
Mean step time (s)	5.781	5.616	5.610	5.616
Median step time (s)	5.62	5.49	5.47	5.47
P90 step time (s)	6.51	6.26	6.32	6.36
Mean TFLOPS/GPU	386.3	397.6	398.1	397.8
Median TFLOPS/GPU	395.85	405.40	407.15	406.50
P90 TFLOPS/GPU	402.8	410.3	413.4	416.6

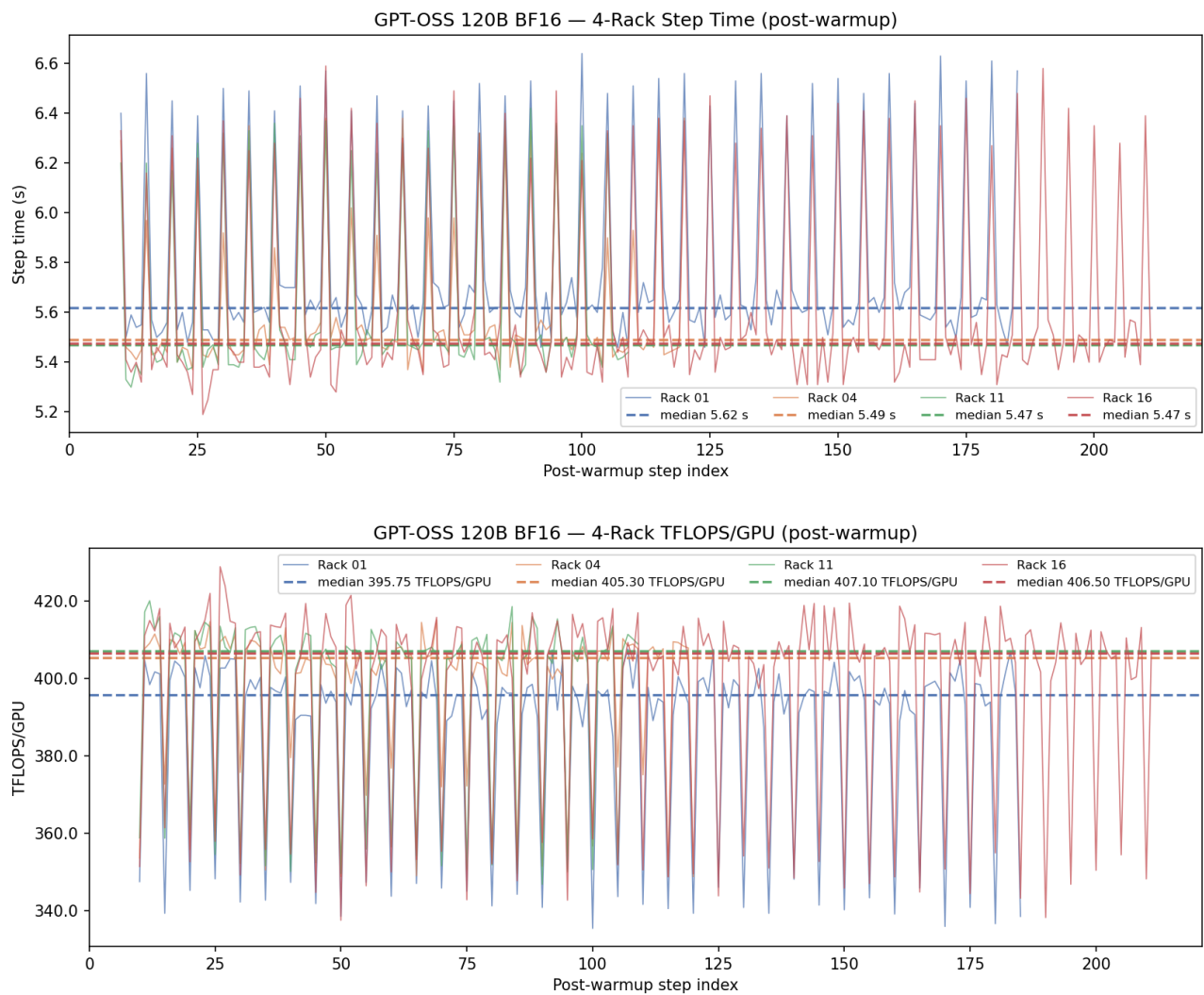
Rack-to-rack deltas (median TFLOPS/GPU, relative to Rack 04 baseline):

Rack	Delta vs Rack 04
01	-2.3%
04	baseline
11	+0.4%
16	+0.3%

Racks 04, 11, and 16 are statistically equivalent (<0.5% spread). Rack 01 is consistently ~2–3% lower throughput across both step time and TFLOPS; this pattern warrants monitoring but is

not indicative of a failure. P90 step time for Rack 01 is also elevated (6.51 s vs 6.26–6.36 s for the other racks), suggesting a small number of elevated steps rather than a uniformly degraded baseline.

Plots



Qwen 72B BF16 — 4-Rack Comparison

Inputs

Rack	Log file
01	logs/prod-cluster/qwen/qwen72b-rack01.log
04	logs/prod-cluster/qwen/qwen72b-rack4.log
11	logs/prod-cluster/qwen/qwen72b-rack11.log
16	logs/prod-cluster/qwen/qwen72b-rack16.log

Results

Metric	Rack 01	Rack 04	Rack 11	Rack 16
Post-warmup steps	61	140	203	79
Mean step time (s)	20.42	20.61	20.61	19.88
Median step time (s)	20.31	20.56	20.57	19.84
P90 step time (s)	21.01	21.18	21.45	20.51
Mean TFLOPS/GPU	714.1	707.4	707.5	733.5
Median TFLOPS/GPU	717.60	709.00	708.70	734.95
P90 TFLOPS/GPU	724.4	719.0	720.8	743.7

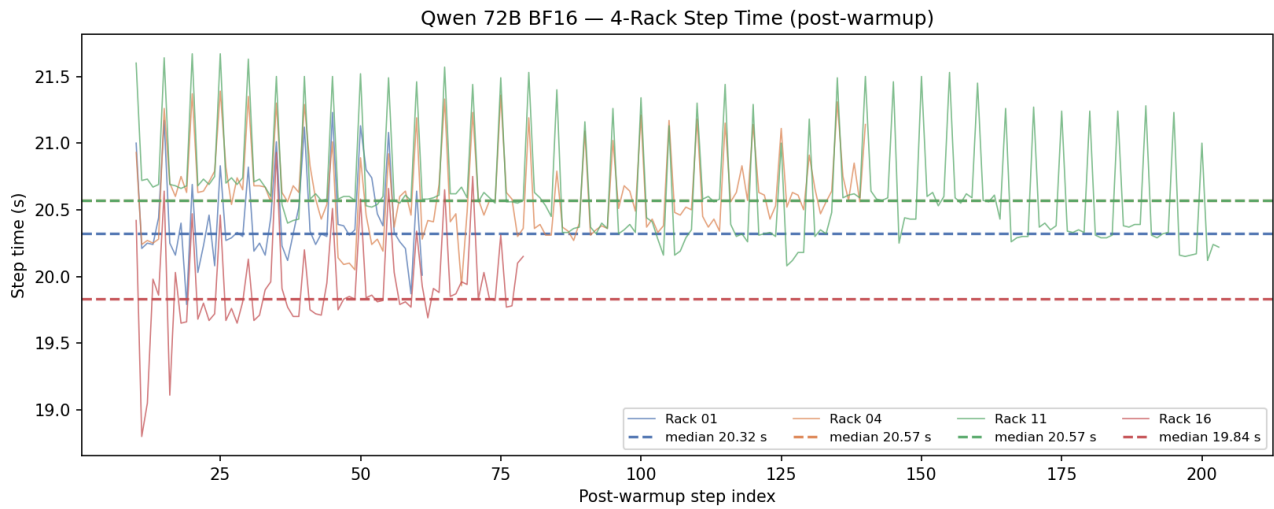
Rack-to-rack deltas (median TFLOPS/GPU, relative to Rack 04 baseline):

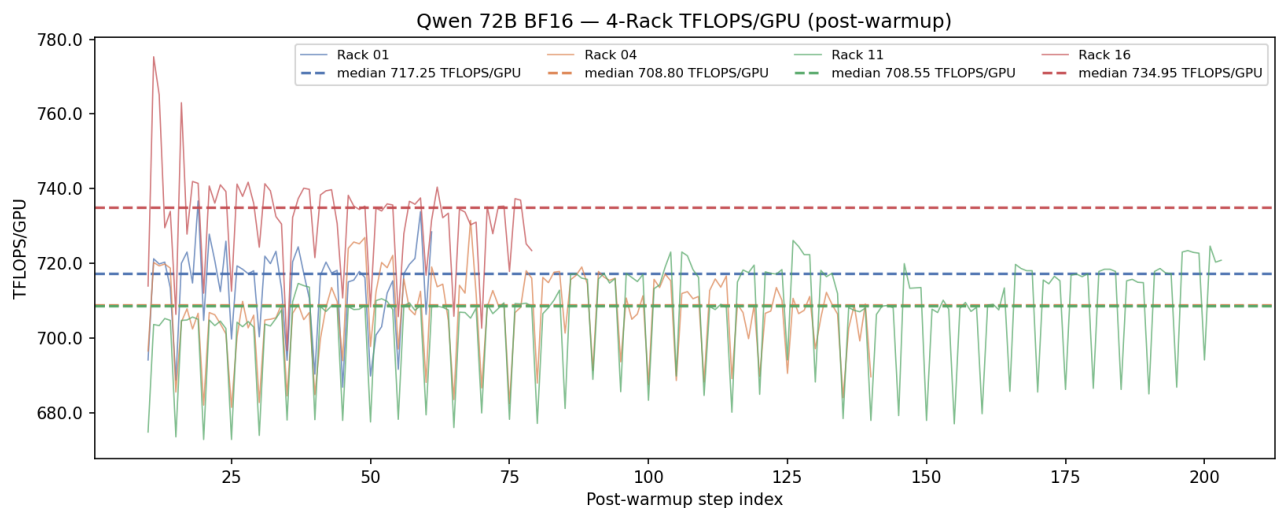
Rack	Delta vs Rack 04
01	+1.2%
04	baseline
11	-0.04%
16	+3.7%

Racks 04 and 11 are effectively identical. Rack 01 is modestly faster. Rack 16 is the strongest performer here, ~3.7% above the Rack 04/11 baseline — notably the inverse of the GPT-OSS pattern where Rack 01 was the weakest. These workload-specific differences are consistent with normal inter-rack NVLink/IB topology variance and do not indicate systematic issues.

Note (v1 continuity): The v1 Qwen 72B comparison used 131 steps per rack (artificially truncated to the shorter log). v2 uses all available post-warmup steps per rack; absolute values may differ slightly from v1 for Rack 04 and Rack 11.

Plots





Nemotron 340B TP8 — Rack Comparison

Both runs use the identical KAI JobSet layout: `dp0 - dp3` (4 replicated jobs × 8 pods = 32 nodes / 128 GPUs), TP=8, PP=4, FP8 `cs` recipe, GBS 128 per-DP group (512 total), KAI scheduler with `nvidia.com/gpu.clique` nodeAffinity pinning.

Inputs

Run	Racks	Log
Rack 04+11	04 + 11	<code>logs/prod-cluster/nemo/nemo340b-kai-longjob-rack4-rack11.log</code>
Rack 01+16	01 + 16	<code>logs/prod-cluster/nemo/nemo340b-tp8-rack1-rack16.log</code>

Configuration (both runs identical)

Parameter	Value
Model	Nemotron4-340B
Parallelism	TP=8, PP=4, CP=1, VP=1, DP=4
MBS	1
GBS (per-DP as logged)	128
Total throughput	512 samples / step
Seq length	4096
Precision	FP8 (<code>cs</code> recipe)
CUDA graphs	enabled
Scheduler	KAI + NVLink clique affinity

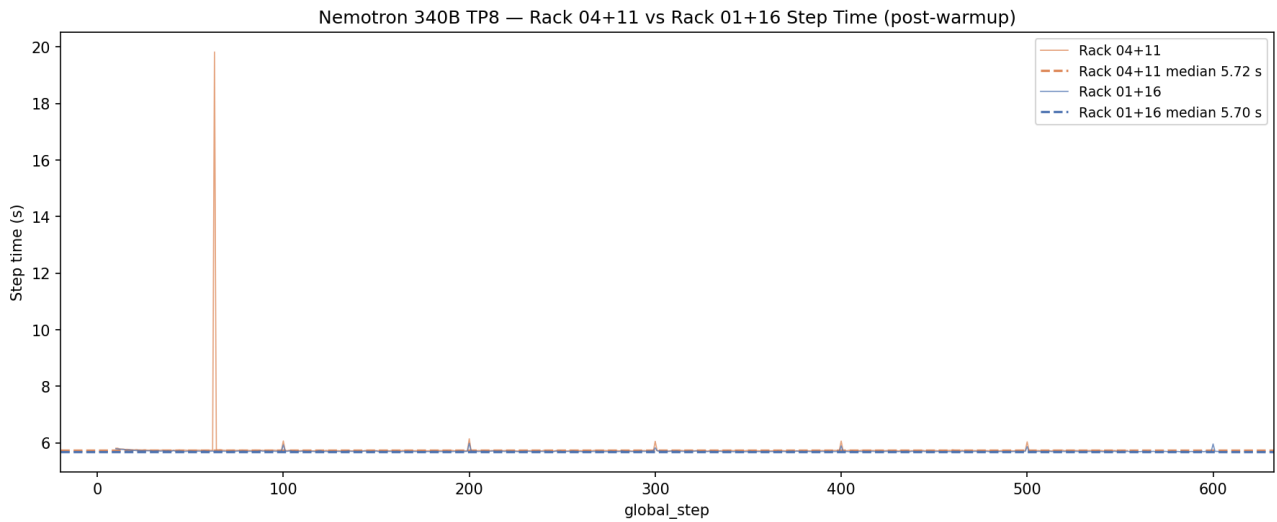
Results

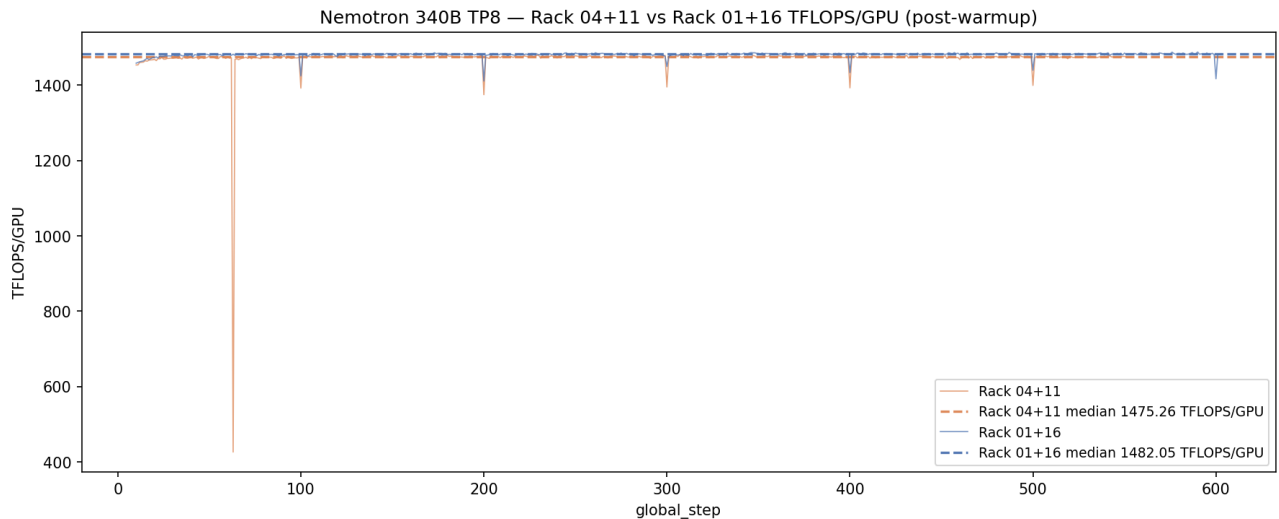
Metric	Rack 04+11	Rack 01+16	Delta
Post-warmup steps	545	594	—
Median step time (s)	5.723	5.697	-0.5%
Mean step time (s)	5.754	5.700	-0.9%
P90 step time (s)	5.737	5.710	—
P99 step time (s)	6.034	5.920	—
Median TFLOPS/GPU	1475.3	1482.0	+0.5%
P90 TFLOPS/GPU	1477.8	1483.3	—
P99 TFLOPS/GPU	1479.4	1485.6	—

Both rack pairs are within normal run-to-run variance (<1%). Rack 01+16 is marginally faster; the slight improvement despite Rack 01 underperforming on GPT-OSS confirms that gap is workload-specific and not a fabric or hardware issue. Either pair is a valid performance baseline for 340B scheduling decisions.

Logging note: NeMo DLL logger emits two entries per step. Statistics use training-TFLOPS entries only (TFLOPS > 100 filter). Rack 04+11 step-63 spike (19.8 s) clipped. First 10 steps excluded as warmup.

Plots





NCCL all_reduce Bandwidth — Rack Sweep

Single-rack NVLS all_reduce sweep (18 nodes / 72 ranks per rack). Out-of-place bus bandwidth (GB/s) reported. All four racks measured.

Message size	Rack 01	Rack 04	Rack 11	Rack 16
128 MiB	476.4	474.9	473.4	475.8
256 MiB	572.5	572.8	574.5	569.8
512 MiB	629.9	632.3	622.1	637.7
1 GiB	737.2	737.2	736.0	737.5
2 GiB	801.9	800.7	803.1	804.3
4 GiB	835.3	833.4	838.6	837.2
8 GiB	875.5	875.7	875.7	877.8
16 GiB	906.1	905.5	908.0	910.4
32 GiB	919.2	918.7	921.6	922.2

Key findings:

- At 32 GiB (saturation), all four racks fall into two tight groups: Racks 01+04 at ~919 GB/s, Racks 11+16 at ~922 GB/s. The spread across all four is **0.4%** — operationally identical.
- Rack 01 and Rack 04 are indistinguishable at every message size. The ~2-3% GPT-OSS training throughput gap on Rack 01 is therefore **not caused by InfiniBand fabric bandwidth**. The cause is workload-specific (e.g. NVLink clique all-reduce path, NUMA binding, or a transient in those log captures). No hardware action required.
- All racks reach >900 GB/s at 16 GiB and plateau cleanly at 32 GiB, confirming the fabric is healthy and fully saturated at large message sizes across the cluster.

Interpretation

- **Cluster-wide health:** All four racks produce within-range performance across all successfully tested workloads. No rack shows a failure signature.
- **Rack 01 GPT-OSS gap (~2-3%):** Consistent across step time and TFLOPS, elevated P90 step time (6.51 s). Not present in Qwen or Nemotron. NCCL all_reduce sweep confirms IB fabric is healthy — Rack 01 and Rack 04 are identical at 919 GB/s at 32 GiB; all four racks within 0.4%. Fabric is not the cause. Likely a software or topology effect specific to the GPT-OSS workload pattern; rerun to confirm before investigating further.
- **Rack 16 Qwen advantage (~3.7%):** Likely reflects a favourable NVLink topology for the Qwen 72B TP=8 all-reduce pattern. No action required.
- **Nemotron 340B TP8:** Rack 01+16 and Rack 04+11 are within 0.5% TFLOPS of each other. Both pairs are production-viable and interchangeable as scheduling targets for 340B jobs.

Source Logs

Workload	Rack	File
GPT-OSS 120B	01	logs/prod-cluster/gpt-oss/gpt-oss-rack01-logs.txt
GPT-OSS 120B	04	logs/prod-cluster/gpt-oss/gpt-oss-rack4-logs.txt
GPT-OSS 120B	11	logs/prod-cluster/gpt-oss/gpt-oss-rack11-logs.txt
GPT-OSS 120B	16	logs/prod-cluster/gpt-oss/gpt-oss-rack16-logs.txt
Qwen 72B	01	logs/prod-cluster/qwen/qwen72b-rack01.log
Qwen 72B	04	logs/prod-cluster/qwen/qwen72b-rack4.log
Qwen 72B	11	logs/prod-cluster/qwen/qwen72b-rack11.log
Qwen 72B	16	logs/prod-cluster/qwen/qwen72b-rack16.log
Nemotron 340B TP8	01+16	logs/prod-cluster/nemo/nemo340b-tp8-rack1-rack16.log
Nemotron 340B TP8	04+11	logs/prod-cluster/nemo/nemo340b-kai-longjob-rack4-rack11.log

Image generation

Charts referenced in this document should be regenerated from the source logs listed above. Suggested filenames (place in logs/prod-cluster/reports/images/v2/):

```
gpt-oss-4rack-step-time.{png,svg}
gpt-oss-4rack-tflops.{png,svg}
qwen72b-4rack-step-time.{png,svg}
qwen72b-4rack-tflops.{png,svg}
```



```
nemotron340b-all-runs-step-time.{png,svg}  
nemotron340b-all-runs-tflops.{png,svg}  
nemotron340b-rack01-rack16-step-time.{png,svg}  
nemotron340b-rack01-rack16-tflops.{png,svg}
```

For GPT-OSS and Qwen charts: one panel per rack, all on shared axes, with per-rack median as a dashed reference line. For Nemotron all-runs chart: four series (NeMo baseline, KAI short, KAI R04+11, KAI R01+16) on shared axes; dashed per-series median lines; note axis break or dual-axis needed since GBS differs.